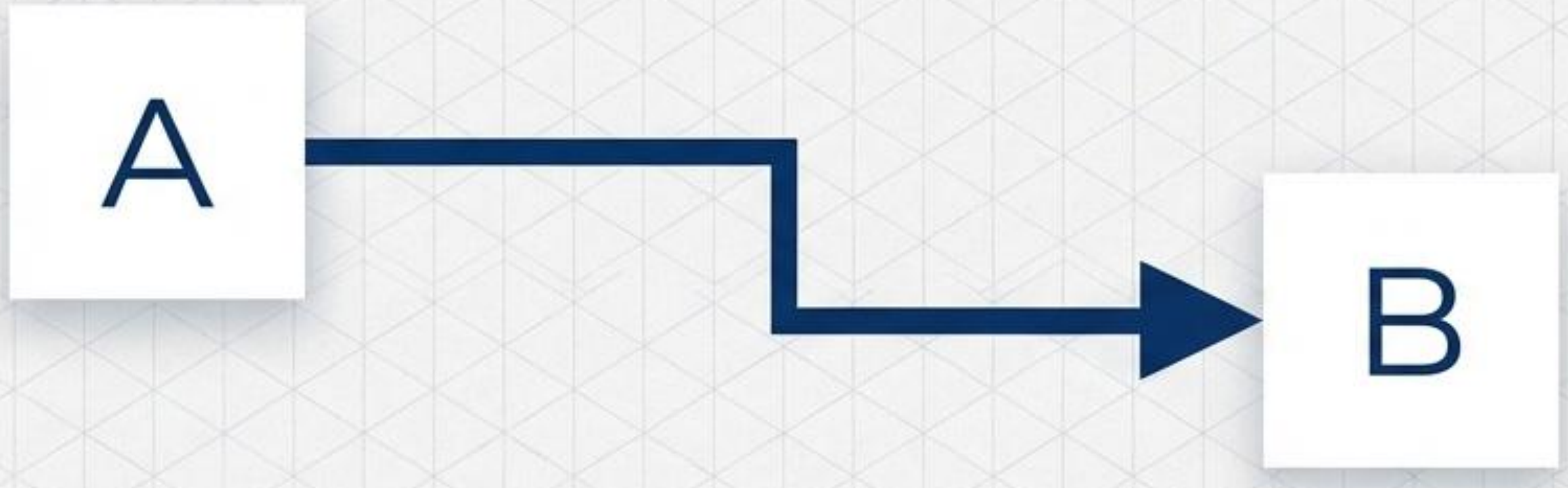
Certyfikat PMP: Project Schedule

dr Zbigniew Galar



Chapter 4: Creating the Project Schedule — Planning Performance Domain

The Weight of the Planning Domain

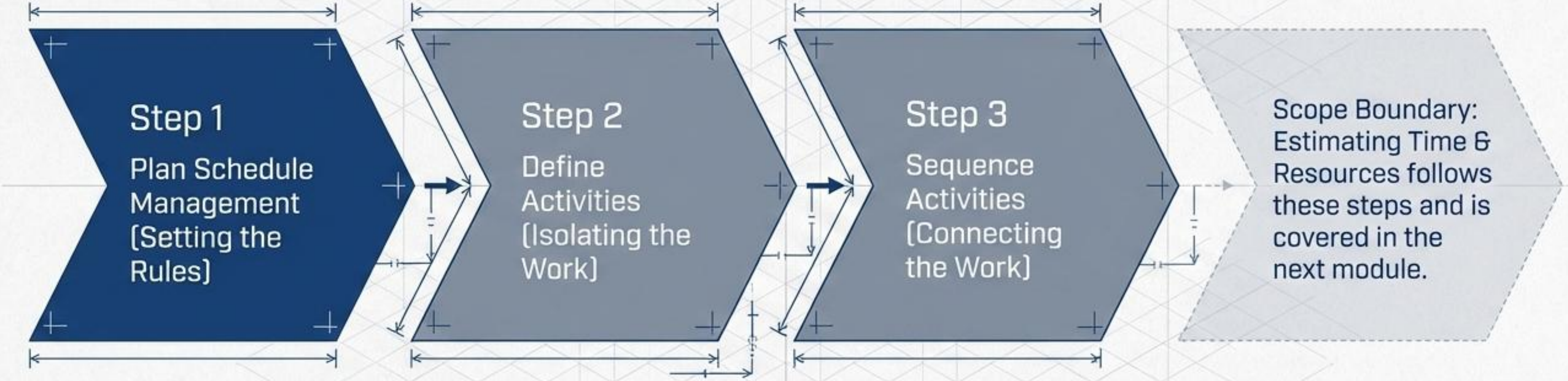
The better planning you do up front, the more likely you'll have a successful project.

EXAM SPOTLIGHT

Planning, Executing, and Monitoring & Controlling process groups account for almost 70% of the PMP® exam questions.

Dedicate your study time accordingly.

Project Schedule Planning Process



Project Schedule Planning Process Overview

Raw Materials (Inputs)

- Project management plan (Scope baseline)
- Project charter
- Enterprise Environmental Factors (EEFs)
- Organizational Process Assets (OPAs)

The Engine (Tools & Techniques)

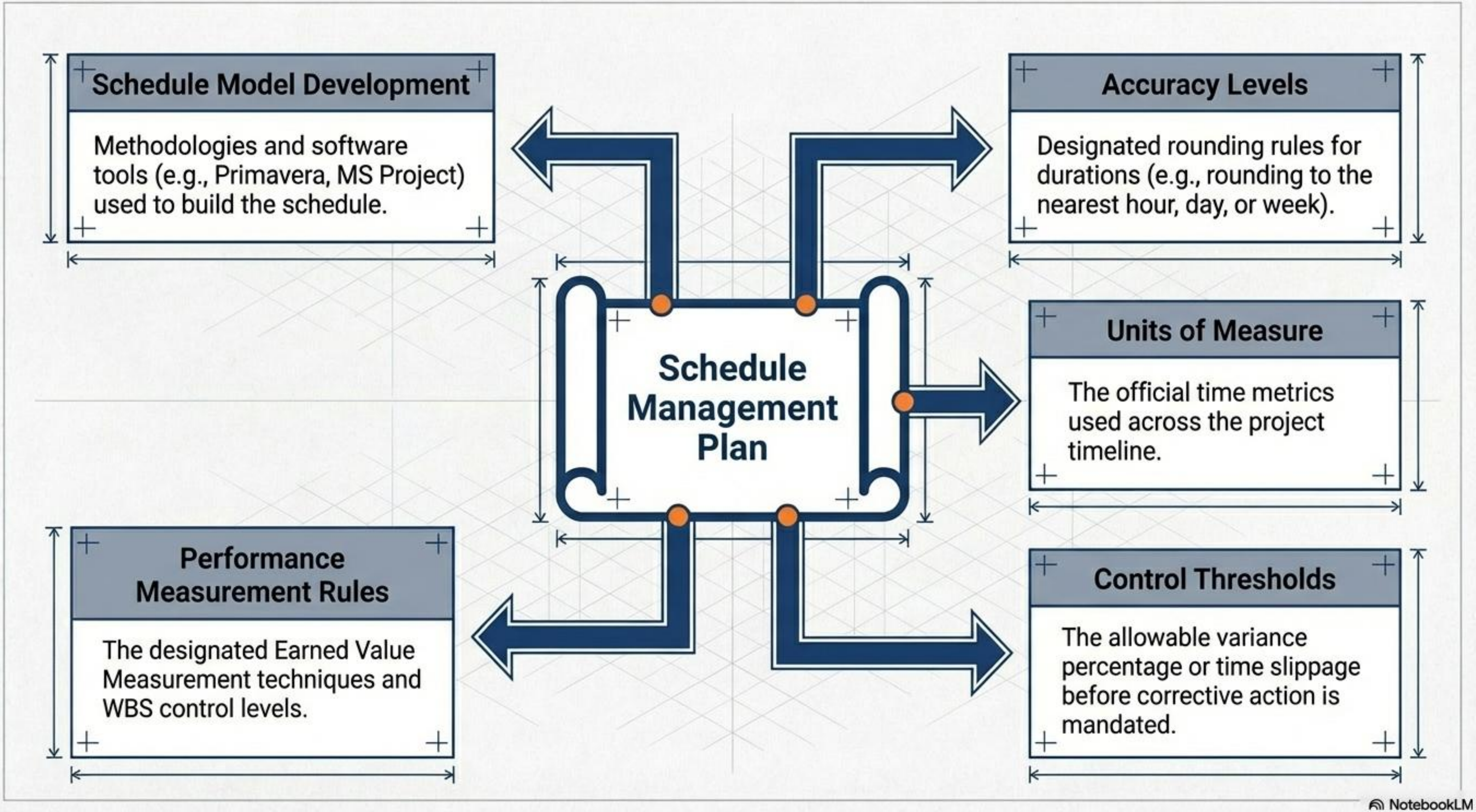
- Expert judgment
- Analytical techniques
- Meetings

Final Product (Outputs)

Schedule Management Plan

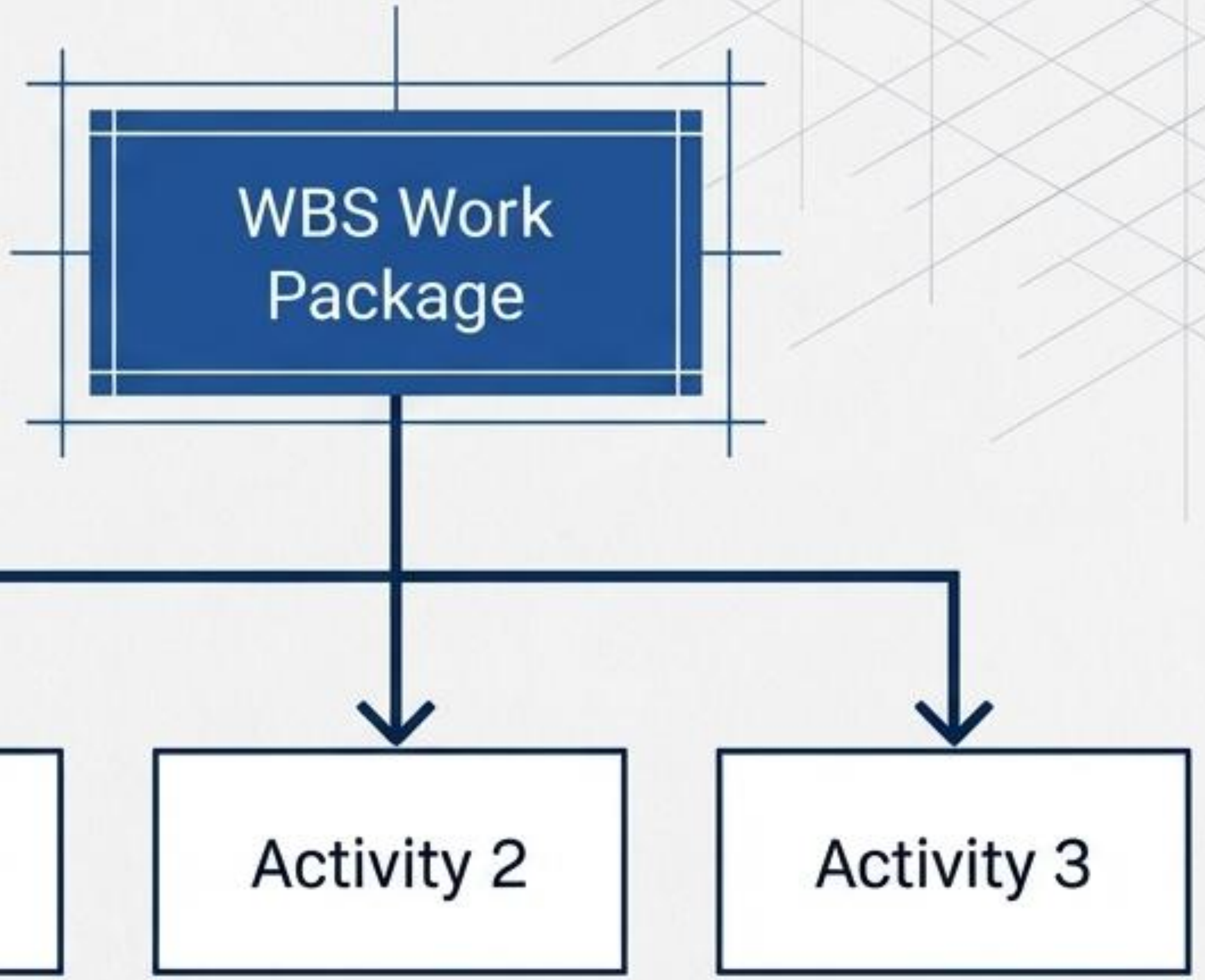
EXAM SPOTLIGHT

The ONLY output of this specific process is the Schedule Management Plan itself.


Schedule Model Development
Methodologies and software tools (e.g., Primavera, MS Project) used to build the schedule.
Accuracy Levels
Designated rounding rules for durations (e.g., rounding to the nearest hour, day, or week).
Schedule Management Plan
Units of Measure
The official time metrics used across the project timeline.
Performance Measurement Rules
The designated Earned Value Measurement techniques and WBS control levels.
Control Thresholds
The allowable variance percentage or time slippage before corrective action is mandated.
NotebookLM

Phase 2: Defining Activities via Decomposition

Breaking down deliverables into the individual, manageable units of work required to fulfill them. This utilizes Rolling Wave Planning—progressive elaboration where near-term work is planned in detail, while future work is planned at a higher level.



EXAM SPOTLIGHT: Activities are the atomic units supporting estimation, scheduling, and execution. They are actions, not deliverables.

The OuThe Outputs of Definition

Activity List

The master index. Contains every schedule activity, a unique identifier code, and a detailed scope of work description.

Activity Attributes

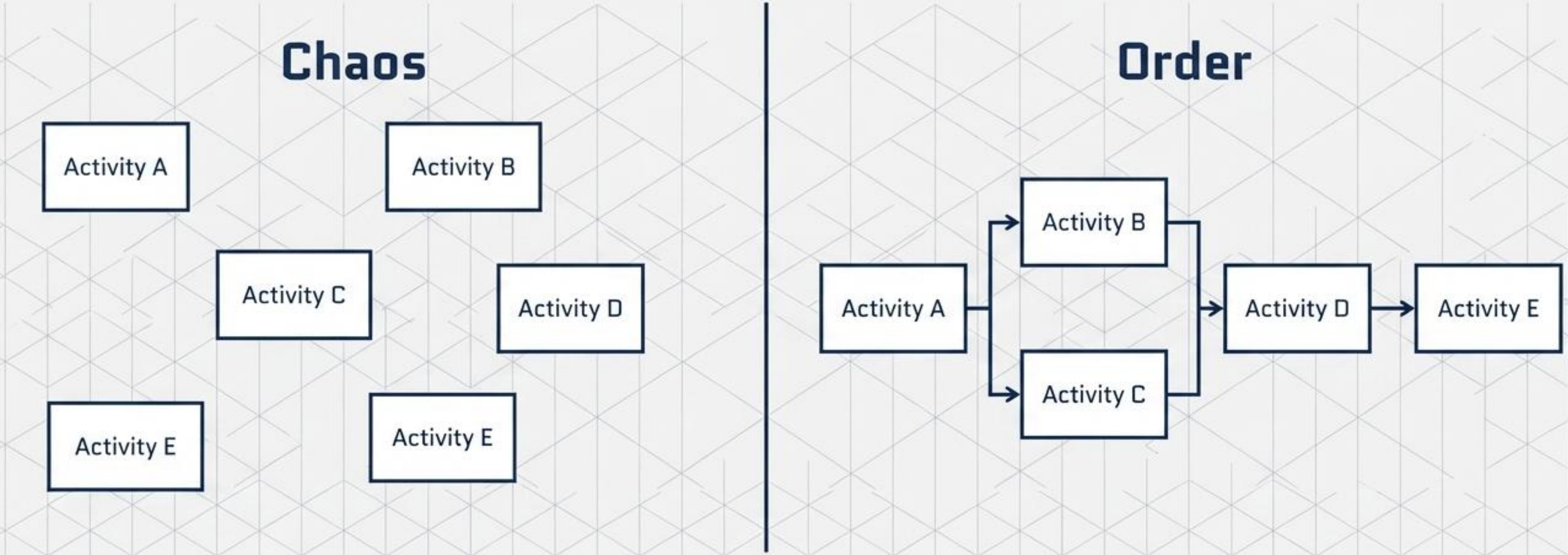
The evolving metadata. Includes predecessor/successor data, WBS codes, logical relationships, leads/lags, and resource constraints.

Milestone List

The major markers. Documents mandatory and optional major accomplishments (e.g., sign-offs, prototypes, system testing).

Phase 3: Sequencing Activities

You have defined the work. Now, interactivity and logical relationships must be sequenced to build an achievable, realistic project schedule. Key tools include Dependency Determination, Precedence Diagramming Method (PDM), and Leads and Lags.



Dependency Determination Matrix: Navigating Logic & Control

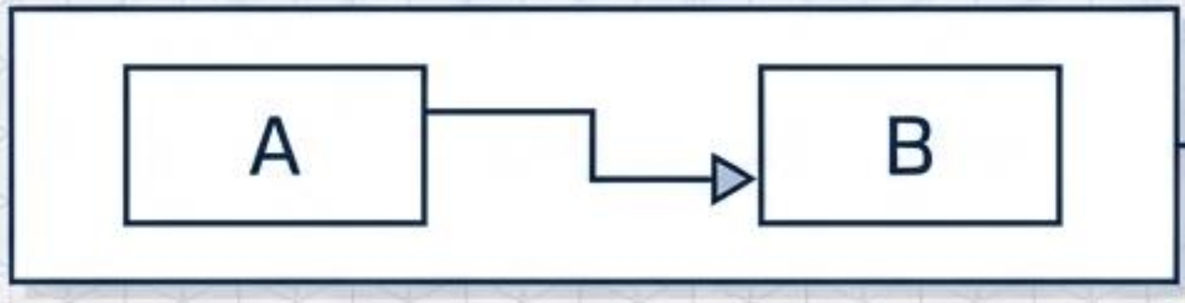
Nature of Logic (Hard/Required vs. Soft/Preferred)

Project Boundaries (Internal Control vs. External Control)

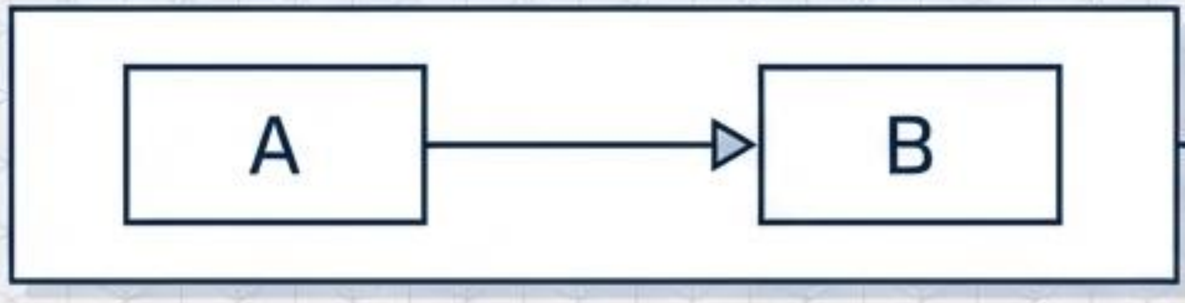
Mandatory Dependencies Physical limitations driven by the work itself (e.g., must build walls before the roof).	**External Dependencies** Non-project activities that block project progress (e.g., FDA approval).
Discretionary Dependencies Best-practice or preferential logic set by the team (e.g., preferring to hand-paint trim).	*(Typically Theoretical)*

EXAM SPOTLIGHT: Discretionary dependencies create arbitrary total float. If you need to compress the schedule (fast-tracking), target discretionary dependencies first for removal or modification!

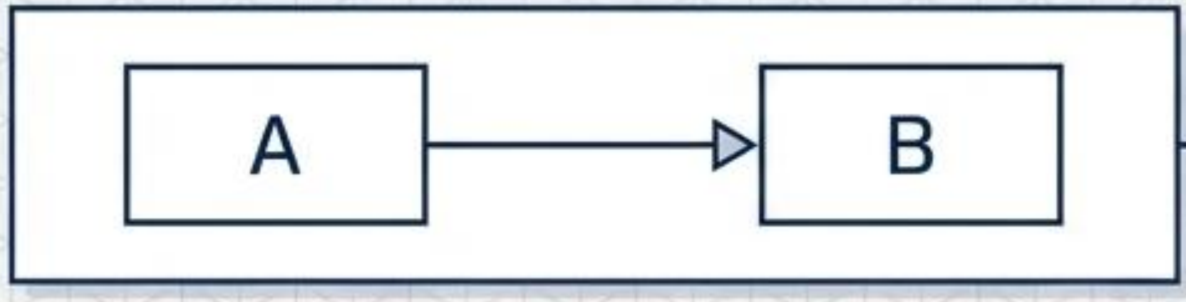
The Four Logical Relationships



Finish-to-Start (FS):
B cannot start until A finishes.



Finish-to-Finish (FF):
B cannot finish until A finishes.



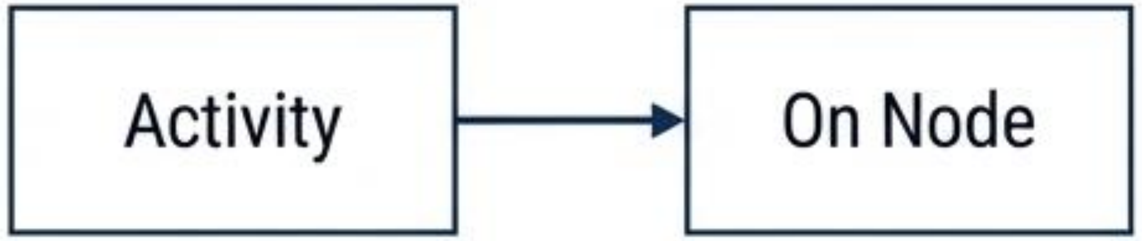
Start-to-Start (SS):
B cannot start until A starts.

Start-to-Finish (SF):
B cannot finish until A starts.

EXAM SPOTLIGHT: Finish-to-Start (FS) is the most common. Start-to-Finish (SF) is rarely used. Every node **must have at least one predecessor** and **successor.**

Diagramminming Methods: PDM vs. ADM

PDM (Precedence Diagramming Method)
Activity on Node (AON)



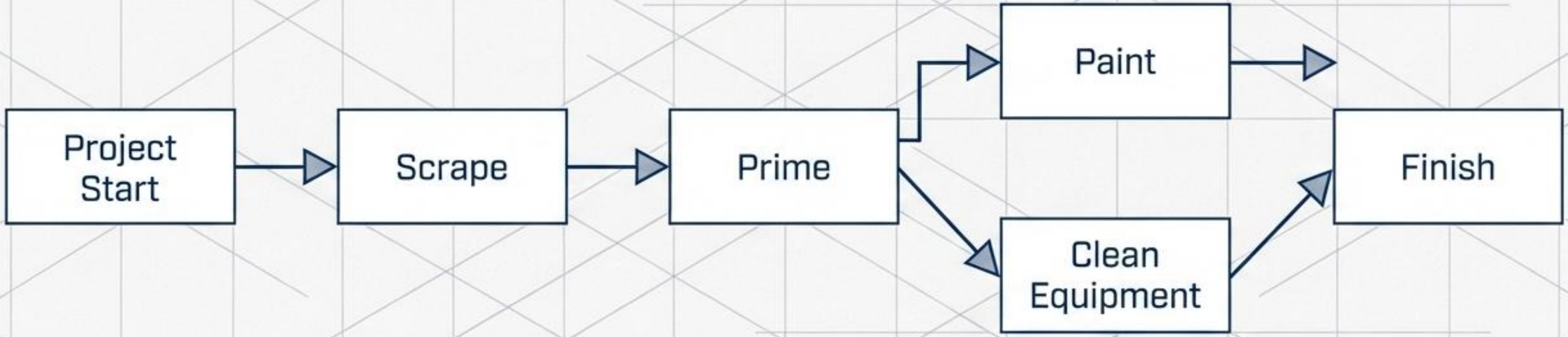
- Visual structure: Activities are boxes; arrows show dependencies.
- Capabilities: Uses all 4 logical relationships. 1 time estimate.

ADM (Arrow Diagramming Method)
Activity on Arrow (AOA)

- Visual structure: Activities are on the arrows; nodes (circles) connect them.
- Capabilities: Uses ONLY Finish-to-Start (FS). 1 time estimate.
- Quirk: Requires Dummy Activities (dotted arrows) to map complex logic.

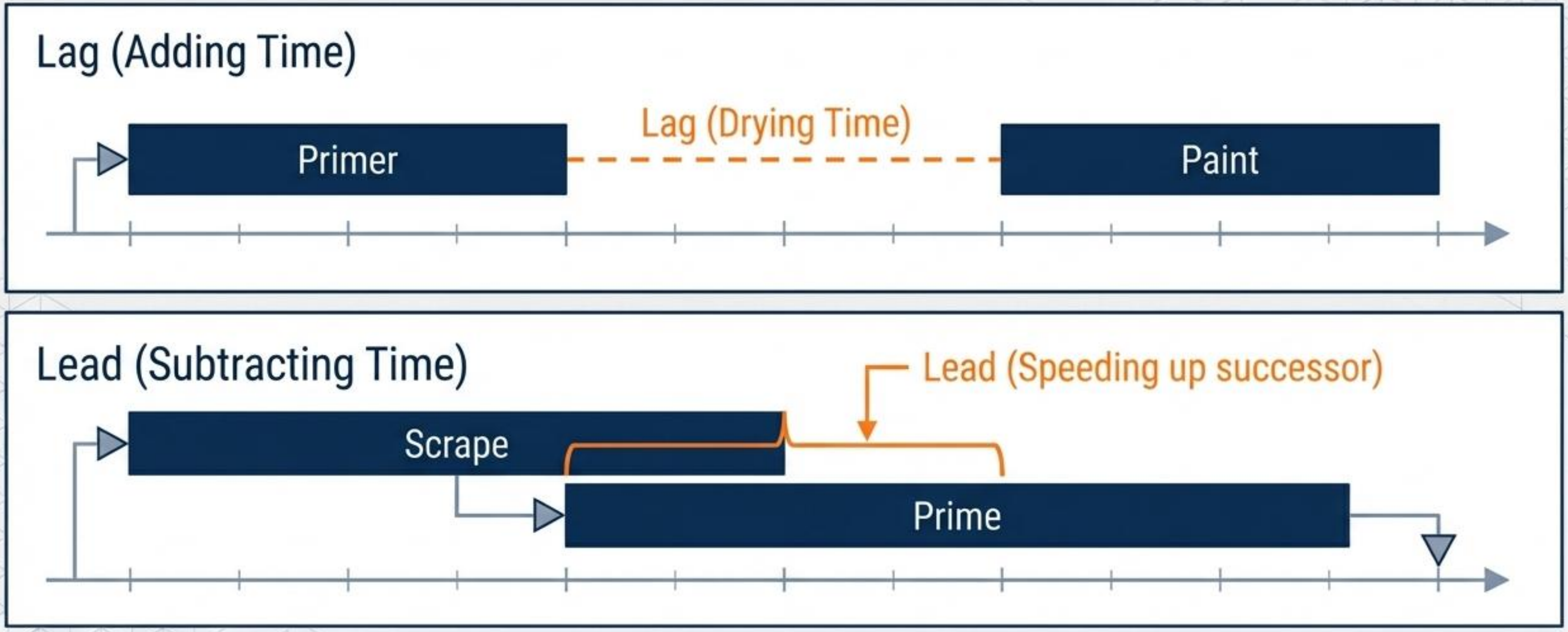
Note on GERT (Graphical Evaluation and Review Technique): Unlike PDM and ADM, GERT allows for conditions, branches, and loops.

Theory into Practice: The Painting Blueprint



Mandatory Dependencies in Action: The physical nature of the work dictates the logic. You cannot prime until scraping is complete. You cannot paint until the primer dries.

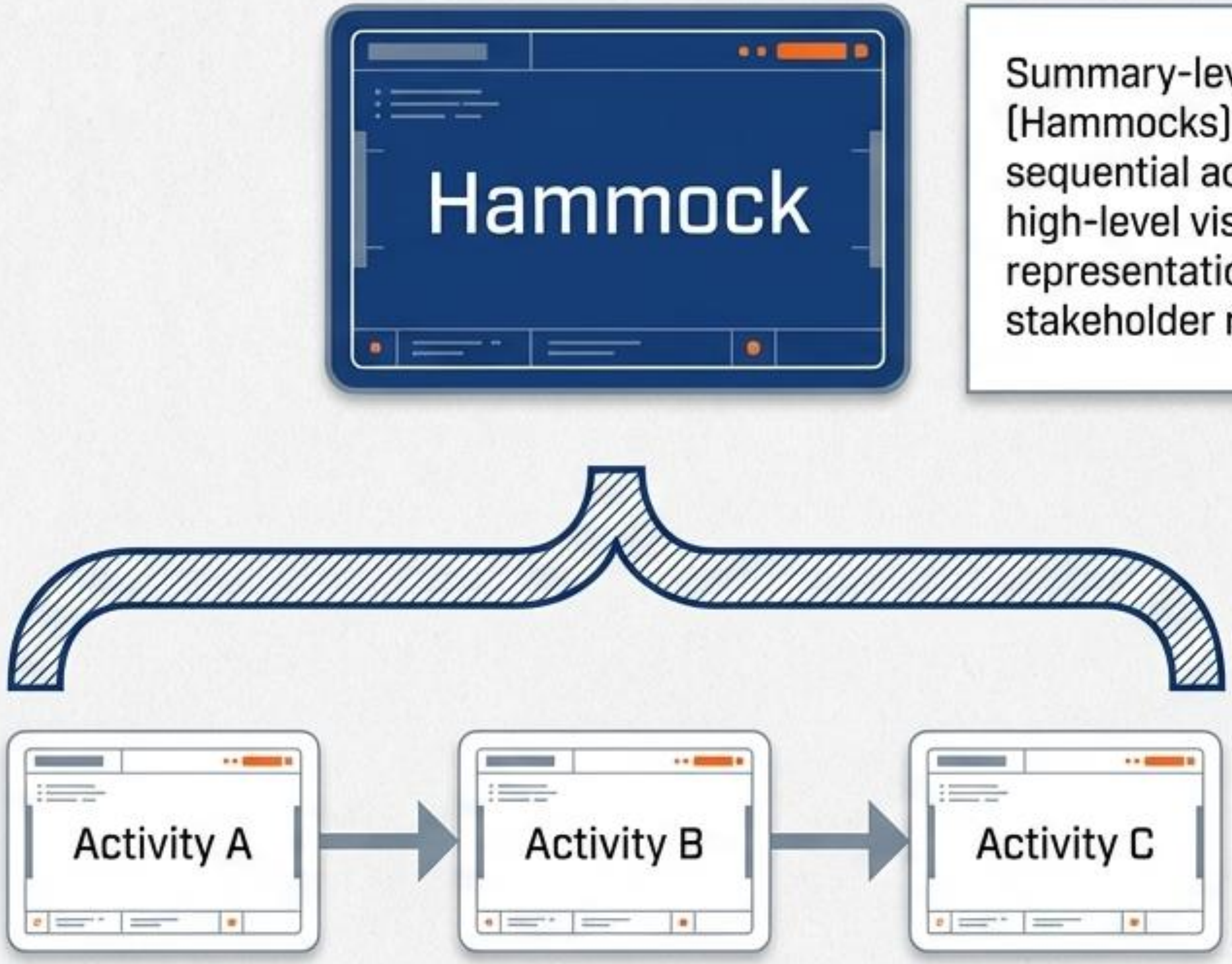
Manipulating Time: Leads and Lags



EXAM SPOTLIGHT: Leads and lags manipulate timing (speeding up or delaying successors), but they must NEVER replace actual schedule logic.

NotebookLM

The Output: Network Diagrams



Summary-level activities (Hammocks) group related sequential activities into a high-level visual representation for stakeholder reporting.

Project Document Updates

The sequencing process often reveals missing steps or reveals missing steps or needed splits. This triggers necessary updates to:

- The Activity List
- Activity Attributes
- The Risk Register

Plan

Defines the rules, software, and thresholds.

Output: Schedule Management Plan.

Define

Decomposes WBS Work Packages into atomic, schedulable actions.

Outputs: Activity List, Attributes, Milestones.

Sequence

Connects actions via logic (PDM/AON) and maps dependencies (Hard/Soft).

Output: Project Schedule Network Diagrams.

The foundation is set and the logic is mapped.
The final step is Estimating Resources and Durations.